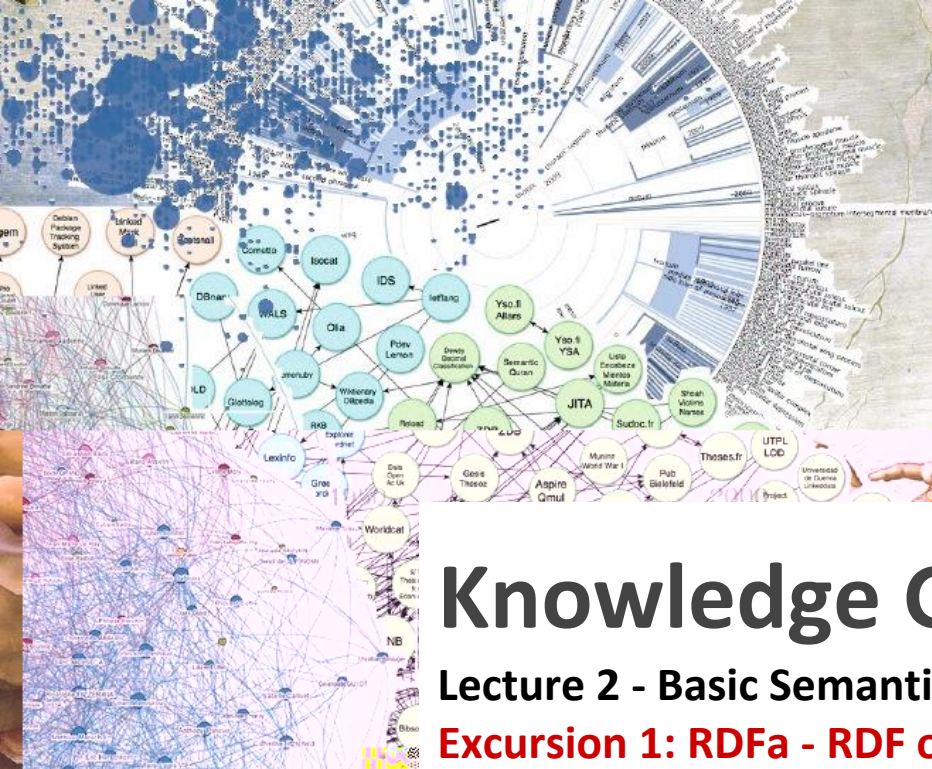




Knowledge Graphs

Lecture 2 - Basic Semantic Technologies

Excursion 1: RDFa - RDF on the Web

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Leibniz-Institut für Informationsinfrastruktur

Knowledge Graphs

Lecture 2: Basic Semantic Technologies

2.1 How to Identify and Access Things

2.2 How to Represent Simple Facts with RDF

2.3 RDF Turtle Serialization

2.4 RDF Complex Datastructures

2.5 Model Building with RDFS

2.6 Logical Inference with RDF(S)

Excursion 1: RDFa - RDF and the Web

The Semantic Web Technology Stack (not a piece of cake...)

Most apps use only a subset of the stack

Querying allows fine-grained data access

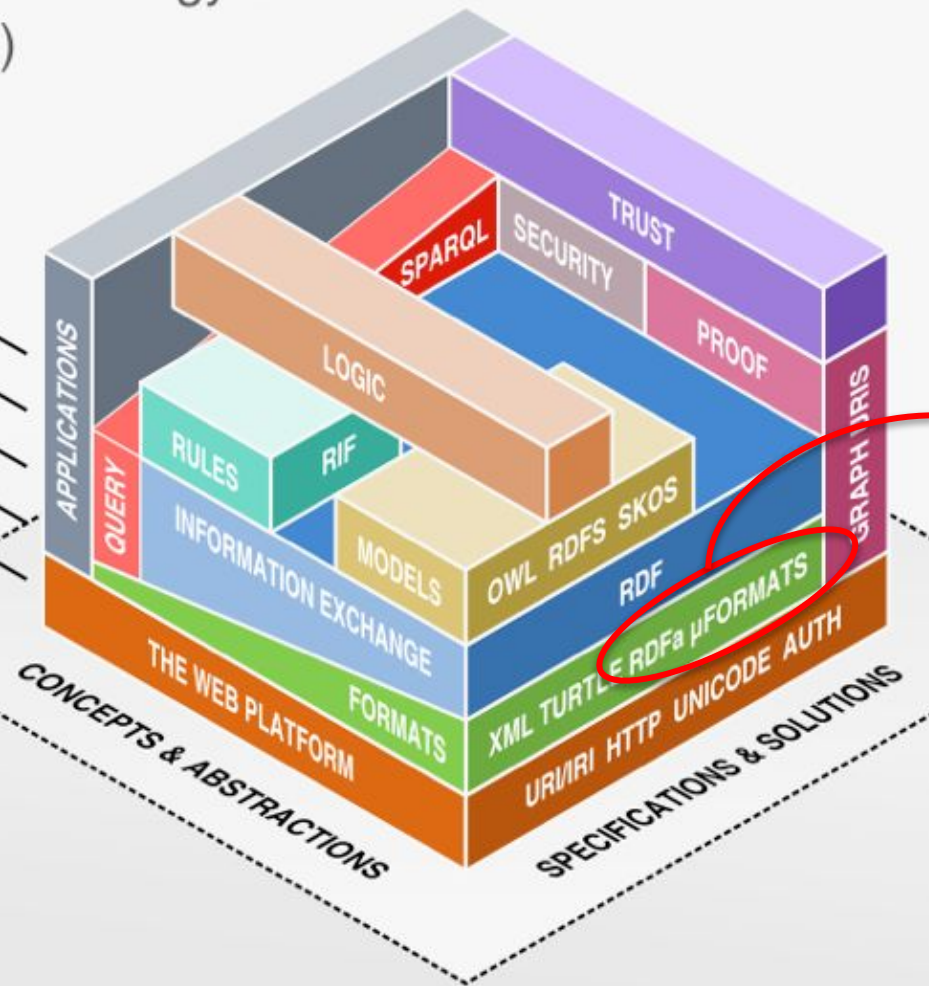
Standardized information exchange is key

Formats are necessary, but not too important

The Semantic Web is based on the Web

Linked Data uses a small
selection of technologies

LINKED DATA



RDFa
and
µFORMATS

RDF(S) Semantics

- In principle there are three ways to embed structured data with explicit semantic annotations within HTML documents.



Domain specific microformats (μ Format)



Generic RDFa



HTML5 Microdata (including schema.org)



Microformats

- Microformats (μ formats) emerged about 2005.
- (X)HTML Markup to express (limited) semantics in an HTML document
 - designed to solve simple, specific problems
 - designed for humans first, machines second
 - used in web pages to describe a specific type of information, as e.g. a person, an event, a product, a review, etc.
- Applications can easily extract data from HTML documents.
- In general, Microformats use the `class` attribute in **HTML tags** (most times `<span>` or `<div>` tags) and assign brief and descriptive names to entities and their properties.

Microformats

- Simple Example: HTML marked up with **hCard** microformat

```
<div class="vcard">
  <span class="fn">Joe Blow</span>
  <span class="title">Senior Manager</span>
  <span class="org">The Example Company</span>
  <span class="adr">
    <span class="street-address">Hauptstr. 123, </span>
    <span class="postal-code">14482</span>
    <span class="locality">Potsdam</span>
  </span>
  Tel.<span class="tel">604-555-1234</span>
  
</div>
```



Microformats - Pros and Cons

- Microformats can **easily be transcoded to RDF** via XSLT.
- New microformat vocabularies **first must be consolidated** by the community, while a new XSLT stylesheet must always be developed for extraction.
- By **using more than one** microformat vocabulary in a single (X)HTML document the processing **complexity increases** rapidly.
- **Conflicts** with used (X)HTML attributes might be possible.



Embedding RDF in HTML Attributes



- RDFa = **RDF** in HTML **a**tttributes
- enables generic RDF annotation in (X)HTML documents by reusing existing (X)HTML attributes.
- RDFa 1.0 based on XHTML (W3C Recommendation 2008)
- RDFa 1.1 based on HTML5 (W3C Recommendation June 2012)
 - **RDFa Lite 1.1**
 - RDFa 1.1



RDFa Lite 1.1

- RDFa reuses existing (X)HTML attributes (e.g. `href`, `src`) and introduces new HTML attributes
 - `vocab`,
 - `typeof`,
 - `property`,
 - `resource`,
 - `prefix`



RDFa Lite 1.1

- First we need a **vocabulary** to talk about things.

```
<p vocab="http://xmlns.com/foaf/0.1/">  
  My name is Harald Sack and you can  
  give me a ring via 1-800-555-0527.  
</p>
```



RDFa Lite 1.1

- Then we have to define the **type of thing** we are talking about.

```
<p vocab="http://xmlns.com/foaf/0.1/"  
  typeof="Person">  
  My name is Harald Sack and you can  
  give me a ring via 1-800-555-0527.  
</p>
```




RDFS Lite 1.1

- Now we can define all **properties** of the thing we are talking about.

```
<p vocab="http://xmlns.com/foaf/0.1/" typeof="Person">
  My name is
  <span property="name">Harald Sack</span>
  and you can give me a ring via
  <span property="phone">1-800-555-0527</span>.
  
</p>
```



RDFa Lite 1.1

- We can create **identifiers** for the things we are talking about.

```
<p vocab="http://xmlns.com/foaf/0.1/"  
  resource="#harald" typeof="Person">  
  My name is  
  <span property="name">Harald Sack</span>  
  and you can give me a ring via  
  <span property="phone">1-800-555-0527</span>.  
    
</p>
```

- `resource` refers to the base URI of the web page.



RDFa Lite 1.1

- And if the vocabulary is not sufficient to describe all properties, we can use additional vocabularies by using prefixes.

```
<p vocab="http://xmlns.com/foaf/0.1/"  
  prefix="ov: http://open.vocab.org/terms/"  
  resource="#harald" typeof="Person">  
  My name is  
  <span property="name">Harald Sack</span>  
  and you can give me a ring via  
  <span property="phone">1-800-555-0527</span>.  
  My favorite beverage is  
  <span property="ov:preferredBeverage">Espresso</span>.  
</p>
```



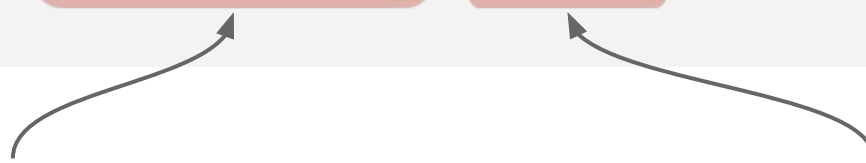
RDFa 1.1

- With full RDFa 1.1 you can add additional functionality:
 - Separate **content** from presentation

```
<p vocab="http://purl.org/dc/terms/">
  <h2 property="title">Syllabus SWT 2012/13</h2>
  <h3 property="creator" resource="#me">Harald</h3>
  Creation Date:
  <span property="created" content="2012-10-28"> 28.10.2012 </span>
</p>
```

content

presentation





RDFa 1.1

- With full RDFa 1.1 you can add additional functionality:
 - Use **datatypes** from XML Schema Definition

```
<p vocab="http://purl.org/dc/terms/">
  <h2 property="title">Syllabus SWT 2012/13</h2>
  <h3 property="creator" resource="#me">Harald</h3>
  Creation Date:
  <span property="created" datatype="xsd:gYear"> 2012 </span>
</p>
```




RDFa 1.1

- Distinguish two different sorts of RDF triples:
 - Triple with **resource** as object
 - Triple with **literal** as object

	Subject	Property	Object
Object is Literal	<code>resource / about</code>	<code>property</code>	<code>content</code> or <code>#PCDATA</code>
Object is Resource (URI)	<code>resource / about</code>	<code>property/rel</code>	<code>href</code> or <code>resource</code>

- `about / rel` only for compatibility with RDFa 1.0



RDFa Tools



<https://www.w3.org/2012/pyRdfa/>

RDFa 1.1 Distiller and Parser

Warning: This version implements [RDFa 1.1 Core](#), including the handling of the [Role Attribute](#). The distiller can also run in XHTML+RDFa 1.0 mode (if the incoming XHTML content uses the RDFa 1.0 DTD and/or sets the `version` attribute). The [package available for download](#), although it may be slightly out of sync with the code running this service.

Distill by URI

Distill by File Upload

Distill by Direct Text Input

Distill by URI

URI:

Output Format:

Returned content:

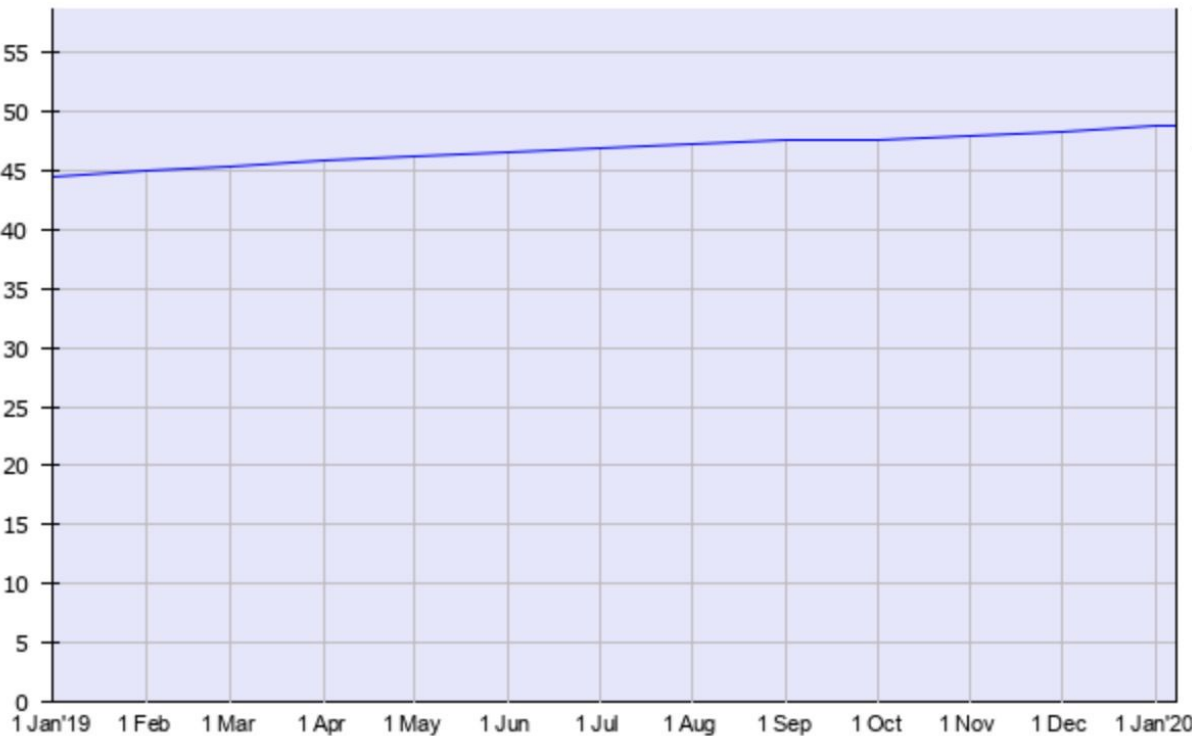
Expand vocabularies:

Generate warnings for non RDFa 1.1 Lite usage:

► [More \(non-standard\) options](#)

Go!

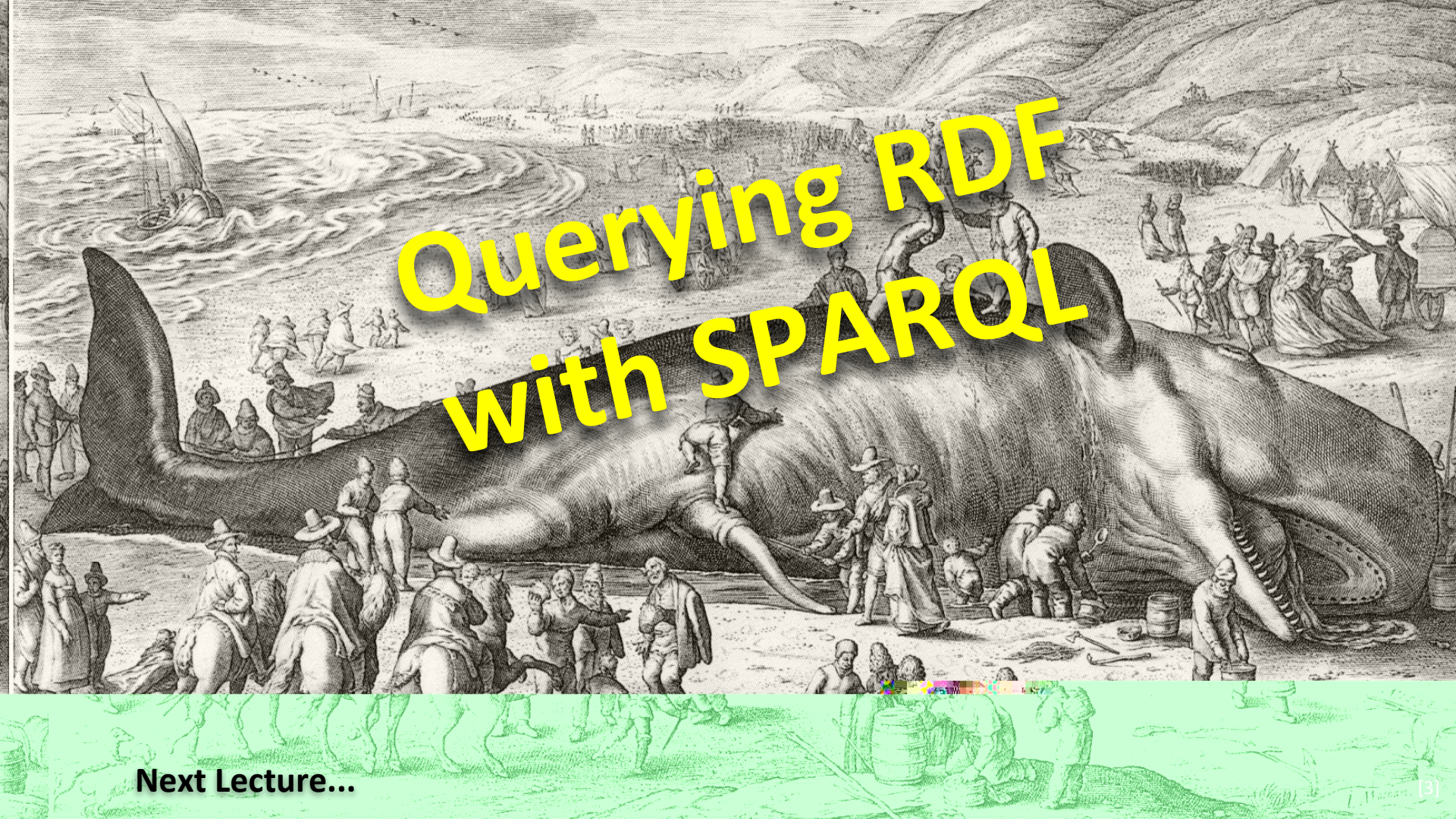
RDFa Usage



Usage of RDFa for websites, 8 Jan 2020, W3Techs.com

Open Graph	94.5%
Generic RDFa	27.7%
W3Techs.com, 8 January 2020	
Percentages of websites using various subcategories of RDFa	
Note: a website may use more than one subcategory of RDFa	

<https://w3techs.com/technologies/details/da-rdfa>



Querying RDF with SPARQL

Next Lecture...

Picture References:

- [1] Benjamin Nowack, *The Semantic Web - Not a Piece of cake...*, at bnode.org, 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>
- [2] Giuseppe Arcimboldo, Vertumnus, a portrait depicting Rudolf II, Holy Roman Emperor painted as Vertumnus, the Roman God of the seasons, c. 1590-1. Skokloster Castle, Sweden. [Public Domain]
https://commons.wikimedia.org/wiki/File:Vertumnus_%C3%A5rstidernas_gud_m%C3%A5lad_av_Giuseppe_Arcimboldo_1591_-_Skoklosters_slott_-_91503.tif
- [3] Hugo de Groot, Stranded Whale at Dutch sea coast in February 1598, [Public Domain]
https://commons.wikimedia.org/wiki/File:Hugo-de-Groot-Nederlandtsche-jaerboeken_MG_0190.tif